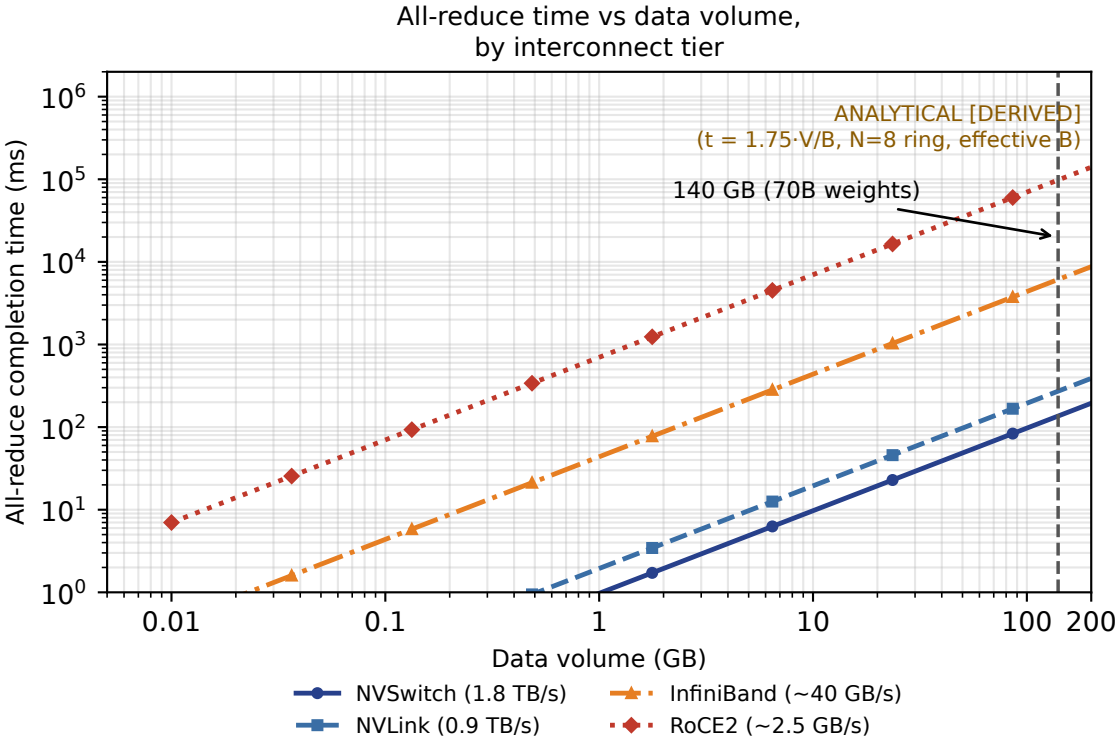
All-reduce time vs data volume,
by interconnect tier
ANALYTICAL [DERIVED]
(t = 1.75·V/B, N=8 ring, effective B)
140 GB (70B weights)
All-reduce completion time (ms)
Data volume (GB)
NVSwitch (1.8 TB/s)
NVLink (0.9 TB/s)
InfiniBand (~40 GB/s)
RoCE2 (~2.5 GB/s)